

Mathematics Review Course

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Swiss Program for Beginning Doctoral Students in Economics 2026

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Course Objectives

Instructor

- Satoshi Fukuda
 - ▶ Econ PhD (UC Berkeley, 2017), Bocconi ('17-'25), Santa Clara ('25-)
- Fields of Research: Pure and Applied (org. econ/political economy/finance) Theory, including Dynamics (both theory and computation)

Goals

- Understand and become familiar with basic mathematical tools for subsequent Econ Ph.D. coursework (Micro, Macro, and Metrics)
- This is a “tool” course.
We stick to learning basic (but abstract) tools (like learning a foreign language)
- My goal is to convey **intuitions** behind the results in both **abstract** (to better understand them **from the higher standpoint**) and **applied** contexts (to use them **in practice**)

Course Topics

Color indicates abstractness

- ① **Probability Theory** (Day 1)
 - ▶ Bayes Theorem in Practice
 - ▶ Conditional Expectations: Theory
- ② **Calculus** and **Optimization** (Day 1–2)
 - ▶ Review of Calculus Techniques
 - ▶ Static Optimization (mainly Lagrangian)
- ③ **Real Analysis** (Day 2–3)
 - ▶ Norm and Distance
 - ▶ Open and Closed Sets
 - ▶ Sequences and Continuous Functions
 - ▶ Completeness of a Metric Space: **Contraction Mapping Theorem**
- ④ **Dynamic Programming** (Day 4)
 - ▶ Theory (Bellman Optimality Principle)
 - ▶ **MATLAB** Implementation through Dynamic Macro-examples

Motivation behind the Selection of Topics

- Using **Bayes theorem** is often harder than you expect $\Rightarrow$ Practice!
- **Conditional expectations** are important in micro/macro/metrics
- **Constrained optimization** (Lagrangian) is important in micro/macro/metrics
- **Projection theorem** is abstract but powerful $\Rightarrow$ as an application we derive the projection matrix
- **Real analysis** is useful to follow your subsequent Econ PhD coursework (in micro/macro/metrics) and papers in economics
- Intertemporal tradeoffs are abundant in economics
 $\Rightarrow$ **Dynamic programming** is useful in **many** fields of economics
- The course structure similar to part of the course I taught at Bocconi & the one that I took at Berkeley

Course Materials

- Lecture Slides (and Codes): self-contained
 - ▶ Obviously I will cover most of the materials, but the lecture slides contain **optional** exercises and applications especially given no problem sets/exam/exercise sessions
- Lecture Notes: as a supplement and/or a reference
 - ▶ E.g., solutions to the optional exercises
- (The reference list in the lecture notes and slides)

Course Expectations

- Generally speaking, the materials covered in this course will be used throughout the first-year Economics PhD coursework and beyond (with varying degree of usage/importance in micro/macro/metrics)
- The ultimate goal is for you to be able to **use** the materials in your coursework and research
- Even if the course materials are a review for you: try to understand them from a higher standpoint
- If some materials are totally new (e.g., real analysis and dynamic programming): you might need time to absorb abstract concepts. Do not rush. The time/effort you spend (and struggles) in this course (including review after the course) would surely help you grasp the materials in your coursework better on during your first year

MATLAB/Octave

- Primary Languages: **MATLAB** or **Octave**
 - ▶ Constrained optimization (Day 2) & Dynamic programming (Day 4)
 - ▶ **MATLAB** is still widely used in macroeconomics; there is a massive library of existing legacy code
- **MATLAB** Access:
 - ▶ Requires a license
 - ▶ Required Toolboxes: “**Optimization**” and “**Statistics and Machine Learning**”
 - ▶ Online version: 20 free hours/month without a license (<https://www.mathworks.com/products/matlab-online.html>)
- **Octave** Access:
 - ▶ Open-source clone (free alternative) of **MATLAB**
 - ▶ Download: <https://octave.org/>
- Course Materials:
 - ▶ Codes provided for both **MATLAB** and **Octave**
 - ▶ **Python** and **Julia** implementations will be briefly mentioned

MATLAB/Octave: Supplementary Resources

(Not needed for the course, and only for your information)

- 1 Solving DSGE Models: Dynare (<https://www.dynare.org/>) runs on **MATLAB**, **Octave**, and **Julia**
 - ▶ Finding the long-run steady state for the system of (non-linear) first-order conditions
 - ▶ (Log-)Linearizing the system of FOCs
 - ▶ Generating impulse response functions
- 2 **MATLAB** Source codes for “Recursive Macroeconomic Theory” (Ljungqvist and Sargent, 2012 3rd (not the latest) edition):
http://www.tomsargent.com/source_code/mitbook.zip,
<https://ideas.repec.org/k/recurse.html>
- 3 QuantEcon (<https://quantecon.org/>): **Python** and **Julia**
 - ▶ MATLAB–Python–Julia cheatsheet:
<https://cheatsheets.quantecon.org/>

MATLAB/Octave: Supplementary Resources

(Not needed for the course, and only for your information)

- ④ Calling **Python** from **MATLAB**, **MATLAB** from **Python**
 - ▶ <https://www.mathworks.com/products/matlab/getting-started/using-matlab-python.html>
- ⑤ Calling **Python** from **Octave**, **Octave** from **Python**
 - ▶ <https://github.com/gnu-octave/octave-pythonic>
 - ▶ <https://github.com/blink1073/oct2py>
- ⑥ Running **MATLAB** or **Octave** on Jupyter Notebook
 - ▶ <https://www.mathworks.com/products/reference-architectures/jupyter.html>
 - ▶ https://github.com/Calysto/octave_kernel

First-Year PhD Coursework (including this course)

General Guide

- Try to understand the **core** ideas by
 - ▶ Reading the course slides
 - ▶ Trying exercises
 - ▶ (Reading the corresponding part of lecture notes)
- Be careful: the course is intense. **Feel free to stop and ask questions**

General Tips

- Study together in a group, including working on problem sets
 - ▶ You'll write papers with your colleagues
 - ▶ Important to be able to discuss research and work together
 - ▶ Even if you do not work on a project with somebody, you will give/receive comments with each other
 - ▶ Friends through your PhD program are going to be your lifelong assets
 - ▶ Make your PhD journey more enjoyable (in a fast-changing world)

Any other questions?

- Any other questions?
- Have you done self-introduction?
 - 1 Your name, PhD Program, Research Interests
 - 2 Any other information you'd like to share here (one sentence)
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